ATTENTION IS ALL YOU N E E D

dominant sequence transduction models ... that include an e
a decoder. The best performing models also connect ... mechanism. We propose
ple network architecture, the Transformer, ... recurrence and convolutio
rely. Experiments on two machine translation tasks ... while being more paralleliza
requiring significantly less time to train. Our model achieves ... English-to-German translation ta
roving over **the existing best results,** including ensembles, by over 2 BLEU. On the WMT 2014 English-to-French translation
model establishes a new single-model state-of-the-art BLEU score of 41.8 after training for 3.5 *days on eight GPUs, a* small frac

he training costs of the best models from the literature. ***Attention*** show that the Transformer generalizes well to other
pplying it successfully to English constituency parsing both with large and limited training data. Recurrent neural networks,
e network architecture, the Transformer, based solely on attention mechanisms, dispensing with recurrence and convolution
rt-term memory **[13] and gated recurrent [VII] neural networks in particular**, have been firmly established ***is*** state of the art
aches in sequence modeling and transduction problems such as language modeling and machine translation [35 , 2 , 5]. Num
rts have since continued to push the boundaries of recurrent language models and encoder-decoder architectures [38, 24, 15]
improving ov**er the existing best results,** including ensembles, by over 2 BLEU. On the WMT 2014 English-to-French translat
models typically factor **computation along the symbol** positions of ***all*** the input and output sequences. Aligning the positi
s in computation time, they generate a sequence of hidden states ht, as a function of the previous hidden state ht−1 and the in
ition t. This inherently sequential nature precludes parallelization within training examples, *which becomes critical at longer seq*
their distance in the input or output sequences [2, 19]. In all but a few cases [27], however, such attention mechanisms are u
gths, as memory ***you*** constraints limit batching across examples. Recent work has achieved significant improvements in co
onal efficiency through factorization tricks [21] and conditional computation [32], *while also improving model performanc*e in
d to their distance in the input or output sequences [2, 19]. In all but a few cases [XXVII], however, such attention mechanis
he latter. The fundamental constraint of sequential computation, however, remains. Attention mechanisms have ***need*** beco